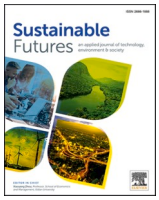
Sustainable Futures 10 (2025) 100984



Contents lists available at ScienceDirect

Sustainable Futures

journal homepage: www.sciencedirect.com/journal/sustainable-futures





Application of machine learning for real-time water quality monitoring in developing countries: A review

Samson Otieno Ooko [a,b,*], Laura Cheptegei [a], Simon M. Karume [a]

[a] Kabarak University, Nakuru, Kenya
[b] Adventist University of Africa, Nairobi, Kenya

ARTICLE INFO

Keywords:
Water quality monitoring
Machine learning
TinyML
IoT (Internet of Things)
Resource-constrained environments

ABSTRACT

Access to clean water is important in ensuring environmental sustainability and public health. However, water quality monitoring in developing countries faces significant challenges, which include; inadequate infrastructure, limited funding, and inadequate skilled personnel. Traditional water quality assessment methods are effective but are time-consuming and require extensive resources, limiting their applicability in developing countries. The emergence of the Internet of Things (IoT) technologies and Machine Learning (ML) present opportunities that can be exploited for monitoring water quality in real-time. Traditional reviews on ML-based water quality monitoring focus primarily on general ML models without considering real-time, low-power implementations in developing countries. To fill the gap, this study systematically reviews the application of Machine Learning (ML) and the Internet of Things (IoT) for water quality monitoring, with a special focus on TinyML's potential for resource-constrained environments. Unlike previous reviews, this study explores model optimization techniques, data limitations, and challenges in scaling ML-based solutions. The key gaps identified include (1) the limited systematic reviews on ML for real-time water quality monitoring, (2) limited integration of TinyML into water quality solutions, and (3) insufficient exploration of the availability of localized datasets. A conceptual model for real-time water quality monitoring is also proposed to address these gaps. This work aims to guide the development of scalable, low-power ML models for water monitoring, especially in developing countries.

1. Introduction

The sustenance of life, especially for humans and animals, depends on access to clean water. However, water quality continues to be a major public health at the global, regional, and local levels. According to WHO estimates, 2.3 billion people lack access to safely managed drinking water services and properly managed sanitation systems and are at risk of water-borne diseases [1]. A recent review of 42 studies from 12 countries in Africa confirmed widespread contamination of surface and groundwater [2]. Surface water contamination was more prevalent at 66.7 %, with groundwater accounting for only 28.6 % of the studies. An increase in pollution in major water sources has been reported in Kenya, leading to growing concerns about water quality in the country [3]. Discharges from urban and industrial wastes, contamination from heavy metals, and agricultural runoff, have been reported as the main sources of water pollution [4,5]. This calls for immediate action to improve water monitoring and management across the globe which will be a step towards the achievement of Goal 6 of the global Sustainable Development Goals (SDGs) [1].

Monitoring of water quality is traditionally done by collecting water samples from different water sources manually of water and analyzing them in a laboratory setting. The samples are often analyzed for different physiochemical and biological parameters that affect the quality of water [2]. Even though the methods used are accurate, they are time-consuming and require the involvement of human experts and equipt laboratories. A substantial amount of investment is also required when extensive field work is needed for data collection. Moreover, given the timeline involved it is not possible to get immediate results that may be needed to respond to water pollution emergencies leading to environmental hazards [6]. Continuous monitoring of water sources in real-time is also not possible with traditional methods, leading to gaps in water quality monitoring data, and affecting the timely water quality management.

When it comes to developing countries, especially in Africa, there are unique challenges associated with water quality monitoring. These countries have inadequate infrastructure, inadequate funding, and a

* Corresponding author.
E-mail addresses: ookosoft@gmail.com (S.O. Ooko), laura.cheptegei@kabarak.ac.ke (L. Cheptegei), skarume@kabarak.ac.ke (S.M. Karume).

https://doi.org/10.1016/j.sftr.2025.100984
Received 2 December 2024; Received in revised form 31 March 2025; Accepted 5 July 2025
Available online 6 July 2025
2666-1888/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

shortage of skilled personnel, making the use of traditional methods of water quality assessments inaccessible [2]. The high costs involved in manual sampling and laboratory testing further worsen the challenges, leading to gaps in water quality monitoring [7]. The emergence of new technologies has led to innovative solutions that can be implemented to help overcome the challenges. IoT and ML technologies have been used to enable remote monitoring and real-time collection of water quality data [8].

Due to the huge computational requirements of existing water quality prediction models, the solutions are cloud-based and require constant internet connectivity to operate [9]. This has limited their applicability to many areas in developing countries where connectivity is usually a problem approach in remote or resource-constrained environments. Moreover, these models are characterized by data transmission delays, and increasing latencies, leading to slow decision-making. The deployment of cloud-based solutions is also faced with privacy and security challenges associated with IoT-driven applications. The application of TinyML is an innovative alternative that can enable the deployment of ML Models on embedded devices in resource-constrained environments [10,11]. It is also easier to scale TinyML-driven models in distributed settings to enable real-time monitoring of water distribution channels.

1.1. Gaps identified

Existing reviews on machine learning for water quality monitoring have primarily focused on conventional ML models that require cloud-based processing, leading to limitations in real-time, on-device implementation in resource-constrained environments. Most studies lack a systematic review that explores the integration of TinyML for low-power, real-time monitoring, which is important for developing countries where connectivity and energy constraints exist. In addition, while previous reviews discuss IoT-based solutions, they do not address the challenges of dataset availability and model validation in real-world conditions. This study bridges these gaps by systematically reviewing the latest advancements in ML and IoT for water quality monitoring, with an emphasis on practical, deployable solutions in developing countries.

1.2. Methodology

The systematic literature review was conducted using IEEE Xplore, ScienceDirect, and Google Scholar, focusing on studies from 2019 to 2024. The keywords used included 'Machine Learning for Water Quality Monitoring,' 'IoT Water Sensors,' and 'TinyML in Environmental Monitoring.' The inclusion criteria were peer-reviewed studies that focus on real-time ML applications for water monitoring, while non-peer-reviewed sources and studies lacking a clear ML application were excluded. The review of existing datasets, methods, tools, and technologies in water quality monitoring was important in identifying the gaps, challenges, and opportunities for future work. The findings of this review will be used as a reference to the development of real-time water quality monitoring models that can be deployed in developing countries.

1.3. Key contributions

The study makes the following key contributions: (1) It identifies the key challenges in applying machine learning for real-time water quality monitoring, particularly in developing countries. (2) It highlights the potential of TinyML for low-power, real-time water monitoring in resource-constrained environments. (3) A conceptual model is proposed for integrating IoT and ML for real-time water quality monitoring. The rest of the paper is organized as follows; water quality datasets are presented in the next section, followed by the ML models used for water quality prediction, the use of IoT in monitoring water quality, and an overview of TinyML, and its application in water quality monitoring. The results are then presented and discussed. A conceptual model for water quality monitoring is then proposed and a conclusion is drawn.

2. Water quality datasets

Data is important in ensuring accurate and reliable water quality assessments. Datasets that contain different water quality parameters make it possible to evaluate and predict water quality by researchers [12]. Some of these parameters include pH, temperature, turbidity, dissolved oxygen, and chemical pollutants. It is easier to get more accurate analyses and predictions with a larger and more comprehensive dataset [13]. With a complete dataset, it is possible to identify patterns and trends that may be difficult to analyze through traditional sampling and laboratory-based methods. An example is the use of multivariate statistical analysis to correlate different water quality parameters and understand their collective influence on water quality [14]. The available datasets have been classified as presented below;

2.1. Traditionally sampled datasets

A dataset is presented in which the data collection method involved collecting 185 water samples from 16 districts in Uganda. The data was collected using sterile containers and sampling from various water sources. The samples were transported on ice to a laboratory from where metal, nutrient, and anion elements were analyzed [15]. The data collection method in another study involved monthly water sampling from 6 stations along two rivers, with in-situ measurements of current velocity, air and water temperature, and laboratory analysis of 16 other physicochemical parameters using standard methods [16]. This same technique was applied in the collection of another dataset, which involved collecting 25 water samples from a stream and its tributaries. This included both urban and suburban areas to assess the water quality based on various index parameters [17]. Moreover, a paper presents a dataset of water pollution data from multiple sources covering over 60 locations in the Ganga basin. The data collection method involved data collection of data from multiple locations using a novel process and lab testing [18]. The processes involved in collecting the presented datasets were time-consuming and involved analysis that took days showing the limitations of using traditional sampling methods.

2.2. Datasets collected by sensors

Sensors have also been used to collect data in different environments. To begin with, Datasets were collected with a wireless sensor network (WSN) with static sensors installed in a river. The sensors measured temperature, electrical conductivity, total dissolved oxygen, and pH, at 10-minute intervals over six months. ZigBee was used to transmit sensor data to a gateway device and then GPRS to transmit the data to a remote server [19]. The use of WSN is a solution to collecting data in open water environments. However, the use of ZigBee may not be applicable in other settings. In addition, this dataset did not capture turbidity, which is also key in water quality monitoring.

A combination of sensor nodes for collecting drinking water quality installed in 24 homes was used to collect 5 water quality parameters. The sensors took measurements every 5 min for 4 weeks. In addition, a periodic manual sampling of free chlorine and lead within each home was also conducted [20]. The combination in the use of both sensors and manual methods is recommended. However, this increased the costs involved.

In addition, multiprobe sensors were used to collect various physiochemical parameters of water. The parameters included dissolved oxygen, turbidity, pH, conductivity, salinity, chlorophyll, temperature, total dissolved solids, and depth of the river. The sensor data was collected every ten minutes and kept in a database using an IoT gateway. Sensor calibration was conducted monthly to the raw data quality was maintained [21]. This dataset was indeed very comprehensive and can

be used for water quality however, it may not be scalable to other areas. In a related solution, sensors installed in municipal water supply systems were used to measure total dissolved solids (TDS), pH, and temperature over some time [22]. In a similar solution, sensors were also used to measure the pH, turbidity, flow rate, and conductivity of water samples in an indoor setting at room temperature over several days [23]. These studies show the capabilities of using IoT sensors in collecting water quality data.

Sensors have also been used in aquatic monitoring, with automated monitoring using digital sensors being used to measure turbidity, temperature, and pH in a fish pond over several days at varying water levels days [24]. In addition, an automated IoT system was integrated with water sensors, which included nitrate, pH, temperature, total dissolved oxygen, ammonia, and turbidity, that collected and uploaded the data in real-time to a cloud platform [25]. Further, autonomous aquatic drones have also been used in data collection. The drone was used to collect sensor data during six water monitoring campaigns. The physicochemical properties of water that were measured in the study included temperature, dissolved oxygen, and electrical conductivity [26]. These solutions show the growing trend in the use of IoT-based sensors for collecting data used in monitoring water quality.

2.3. Remote sensing and satellite data

Remote sensing has also been used in collecting water quality data. A dataset was created with satellite-based spectral reflectance data from the Landsat satellites using custom-built tools [27]. In another study, the water quality data was also paired with tripod-mounted hyperspectral imagery, synchronous proximal spectra, and satellite data [28]. These studies show the capabilities of using satellites for data collection in this area of study.

Table 1 gives a summary of the reviewed water quality datasets showing the data collection methods, area of collection, and parameters measured.

3. ML models for predicting water quality

It was noted in a review on the use of machine learning for predicting water quality that ML is an important tool for analyzing, classifying, and predicting water quality in different water environments [30]. This has been supported by a review that noted that ML could improve water quality monitoring by enabling forecasting and early detection of water contamination [31]. Given its capabilities, machine learning has been adopted widely for analyzing and predicting the quality of water [32]. The studies analyze the physicochemical properties of water to classify water as safe or unsafe for domestic use [33]. These reviews show the capabilities of machine learning in predicting the quality of water. The ML models have been grouped into different categories, as presented in the following section.

3.1. Traditional machine learning models

Traditional ML algorithms have been used in predicting drinking water quality with success [34]. In this study, the K-Nearest Neighbor (KNN) and Support Vector Machine (SVM) had the best performance when evaluated. In a related study, a model was developed for predicting water quality from pH, temperature, and dissolved oxygen [35]. In addition to KNN and SVM used in the previously highlighted studies, Decision Tree, Naive Bayes, and Random Forest algorithms were also used in this study. The model performance was evaluated based on the accuracy, analysis of the confusion matrix, precision, recall, and F1-score.

Other studies have also presented similar trends through the application of ML for predicting water quality based on the physiochemical properties of water [36,37]. The ML algorithms applied in the studies also included Naive Bayes, KNN, Decision Tree, Random Forest, and

Table 1
Summary of water quality monitoring datasets.

Dataset Title	Collection Method or Device	Area of Collection	Parameters Measured
Uganda Water Samples [15]	Collected 185 water samples using sterile containers; transported on ice to a lab for analysis	Various water sources in 16 districts of Uganda	Metal, nutrient, and anion elements; dissolved oxygen; pH; electrical conductivity; turbidity
Integrated Global Water Quality Data [29]	Integrated existing data from five global/regional sources (CESI, GEMStat, GLORICH, Waterbase, USGS WQP)	Global	Various water quality parameters (unspecified)
Ubogo and Egini Rivers Sampling [16]	Monthly water sampling from 6 stations; in-situ measurements and laboratory analysis using standard methods	Ubogo and Egini rivers	Current velocity; air and water temperature; 16 physicochemical parameters
Neeru Stream Water Quality Assessment [17]	Collected 25 water samples; assessed water quality based on index parameters	Neeru stream and its tributaries (urban and suburban areas)	Various water quality index parameters
Ganga Basin Water Pollution Data [18]	Combined authors' own data collected via a novel process with previously inaccessible lab results	Over 60 locations in the Ganga basin, India	Water pollution data (parameters not specified)
WSN in Sitnica River [23]	Wireless sensor network with static sensors; measurements every 10 min; data transmitted via ZigBee and GPRS	Sitnica River	Temperature; electrical conductivity; pH; dissolved oxygen
Sensor Nodes in Detroit Homes [20]	Sensor nodes installed in 24 homes; measurements every 5 min for 4 weeks; periodic manual sampling	Homes in Detroit, USA	Five drinking water quality parameters; free chlorine; lead
Multiprobe Sensors in Chao Phraya River	Eureka Manta+35 multiprobe sensors; measurements every 10 min; data stored via IoT gateway; monthly calibration	Chao Phraya River, Thailand	Total dissolved oxygen; salinity turbidity; conductivity; pH; total dissolved solids; temperature; chlorophyll; depth
Municipal Water Supply Monitoring [22]	Sensors installed in municipal water supply systems; data collected over time	Municipal water supply systems (location not specified)	Total dissolved solids; pH; temperature
Indoor Sensor Measurements with Contaminants	Sensors in an indoor setting; introduced different contaminants over several days	Laboratory setting	pH; turbidity; flow rate; conductivity
Digital Sensors in Fish Pond [26]	Continuous automated monitoring using digital sensors at two water levels over multiple days	Fish pond (location not specified)	Temperature; pH; turbidity

(continued on next page)

Table 1 (*continued*)

Dataset Title	Collection Method or Device	Area of Collection	Parameters Measured
Automated IoT Water Quality System	Automated IoT system with water quality sensors; real-time cloud data upload	Not specified	Temperature; pH; dissolved oxygen; turbidity; ammonia; nitrate
Autonomous Aquatic Drones Data [26]	Autonomous aquatic drones (Platypus Lutra boats); data collected during six campaigns in both autonomous and manual modes	Not specified	Temperature; dissolved oxygen; electrical conductivity
Ground Measurements with Landsat Data	Paired ground-based water quality measurements with satellite spectral reflectance data from Landsat	Not specified	Total suspended sediment; dissolved organic carbon; chlorophyll-a; Secchi disk depth
Water Quality Data with Hyperspectral Imagery [27]	Collected over 200 datasets from 2016 to 2018; lab analysis and in-situ sensors; paired with hyperspectral imagery and satellite data	Rivers and Eight lakes USA	chlorophyll-a; Blue-green algae phycocyanin; specific conductivity; dissolved oxygen; pollution sediments turbidity;fluorescent dissolved organic matter

SVM. Unlike the previous studies, the performance of the models was improved by hyperparameter tunning. The performance enhancement approach was also used in a related study in which a lightGBM model was used in addition to Decision Tree, KNN, SVM, and Random Forest, with the best accuracies being recorded after hyperparameter tunning. These studies confirm the possibilities of improving the performance of models for predicting the quality of water.

Multi-layer perceptron (MLP) and Naïve Bayes were also applied in related studies to accurately predict the quality of water. In this study, the MLP algorithm achieved the best accuracy compared to Naïve Bayes [38]. This shows the potential of exploring additional models in the prediction of water quality.

From an analysis of the studies that applied traditional ML models for predicting water quality, it was noted that the Random Forest classifier was the best-performing model providing better accuracies compared to other models, as presented in [39] and [40]. This finding is also supported by a study that used supervised learning models in which the Random Forest algorithm performed better than other models used in the study [41]. The researchers therefore recommend the use of Random Forest when applying the traditional ML-based models for prediction of water quality.

3.2. Advanced machine learning models

Advanced ML techniques, including Artificial Neural Networks (ANN), have also been applied in predicting water quality. In a recent study, a model for predicting water quality using ANN and time-series data is presented [42]. The study confirms that ANN can be used to classify complex environmental datasets with the capability of capturing the non-linear relationships in the data. The models have also been used for Water Quality Index (WQI) classifications with parameters being optimized [43]. The classification models used in the study include RF, Extreme Gradient Boosting (Xgboost), Gradient Boosting (GB), and Adaptive Boosting (AdaBoost) with the GB model achieving the best performance in terms of accuracy in predicting the WQI. In addition, Regression models were also used in this study, and they included the KNN regressor, DT regressor, SVM Regressor, and MLP regressor. The MLP regressor model performed better than the other regression models in predicting the WQI.

3.3. Prediction based on physiochemical parameters

ML models are effective in predicting the quality of water based on physiochemical and microbiological parameters [44]. An ANN model was applied to predict the quality of drinkable water from the physiochemical properties of water [45]. A study that integrated ML and IoT-based sensors confirmed that the integration of these technologies improved the accuracy of water quality assessments as compared to the traditional methods [12]. The ML models used in the study include SVM, Deep Neural Networks (DNN), and KNN. ML was also used to predict river water quality in [46]. These models used in the study include SVM regression, DT, ANNs, and LSTM networks, which is a deep learning model. An integrated framework integrating IoT and LM for analysis and prediction of water quality is also proposed [47]. In this study, Regression models, including LSTM, SVR, MLP, and NARNet were used to predict the WQI with the MLP regression model giving the best performance. Classification models including SVM, XGBoost, DT, and RF were used to predict the Water Quality Class (WQC) with the RF classification model giving the best performance.

ML models have also been used in predicting the quality of irrigation water from electrical conductivity and pH [48]. The ML models used in the study were ANN, MLR, SVR, KNN DT, RF, Stochastic Gradient Descent (SGD), and AdaBoost. the SVR and KNN models were the worst-performing models. In another study, MLR and RF were the best-performing model in predicting WQI [49] with SVM being the worst-performing model. In a related study XGBoost, DT, AdaBoost, KNN, and SVM were used to predict water quality and also give a projection of water quality in the future [50].

3.4. Use of remote sensing in predicting water quality

In a review of the application of machine learning algorithms for predicting water quality parameters using remote sensing data [39], it was noted that ANN and SVR models performed better than other models. The adoption and use of remote sensing technologies for water quality monitoring is however still limited and require further investigation.

Satellite-based water quality monitoring using ML algorithms is also a promising approach to addressing water the water quality problem [51]. Neural networks (NNs) and deep neural networks (DNNs) are the most popular and widely used models and the give better performance as compared to other models. Linear regression models have limitations in capturing complex non-linear patterns in the data. The use of ML models based on satellite data for predicting water quality was further presented in a review [52]. This study reviewed studies that used ANN, RF, SVM regression, cubist, genetic programming (GP), and DT.

Fig. 1 gives a mind map of the application of ML in water quality monitoring.

4. Water quality monitoring using IoT

From recent review studies, it was noted that IoT increasingly being applied to water quality monitoring in different environments [53,54]. The IoT sensors are cheap and can measure pH, temperature, turbidity, TDO, TDS, and electrical conductivity with microcontrollers being used to integrate and interface with the sensors. Wi-Fi, cellular networks, LPWAN technologies, Bluetooth are used for connectivity for remote monitoring and data transmission to the cloud for analysis and real-time decision-making.

An IoT-based smart water quality assessment framework for aquaponds management was recently developed [55]. This solution uses, different water quality sensors including pH, DO, temperature, turbidity, ammonia, nitrate, and manganese sensors. WiFi was used for sending data to the Cloud-based storage on Google Drive. This shows the

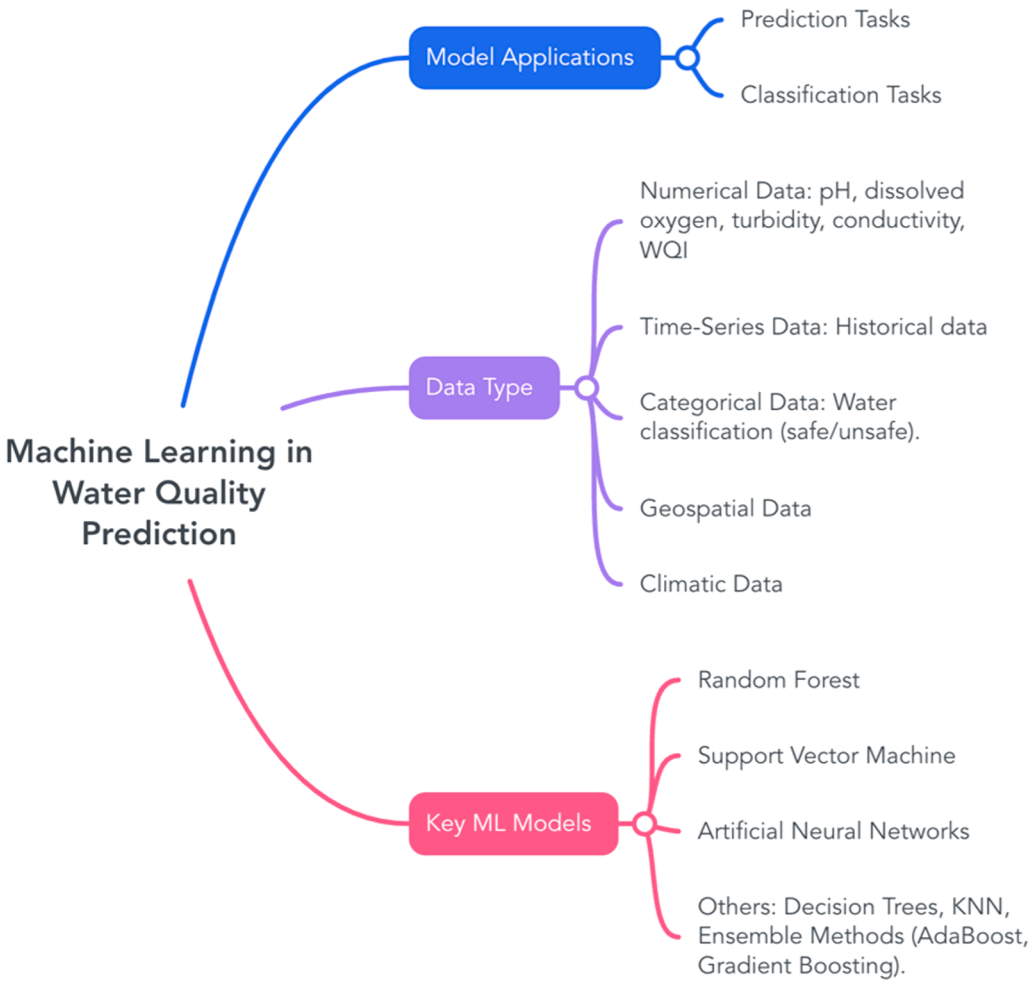


Fig. 1. A mind map of the use of ML in water quality monitoring.

ability of IoT sensors to collect water quality data. This is further confirmed by a related study that presents an IoT-based water quality monitoring system for aquaculture. The solution measures key parameters including dissolved oxygen (DO), pH, and temperature in real time [56]. An Arduino Uno board is used as the microcontroller with a GSM communication module to send data to the cloud server for storage.

The IoT driven solutions have also ben applied in monitoring water bodies. A study presents an IoT driven solution for monitoring the quality of water in a river. This is done using sensor to collect data followed by a statistical analysis to calculate the WQI which is then used to classify the water quality for different usages [57]. The sensors used in the solution include pH, dissolved oxygen, conductivity, ORP, and temperature. A gateway is used to collect data from Waspmotes and send the collected data to the cloud using including Bluetooth, WiFi, and 4 G [57]. In a similar solution, a Low-cost IoT drvien solution has been proposed for monitoring the qaulity of lake water. This model is a solar-powered with the capability of monitoring of five water quality parameters. The monitored parameters include total dissolved solids, turbidity, dissolved oxygen, pH, and temperature. An Arduino board is used as a microcontroller with a Wi-Fi module sending the collected data to a web server and a Micro-SD card to store sensor data locally.

There have been attempts to design IoT-based solutions tailored for deployment in rural areas. To begin with, a solution for monitoring water quality using IoT in rural areas is presented [58] The study presents the development of a low-cost IoT-based system for monitoring water quality in rural areas. In the solution, an Arduino is used as a microcontroller with temperature, turbidity, pH, and TDS sensors being used for collecting data. The users used a mobile application to view the collected data and recommendations with Bluetooth as the communication protocol. In addition, In a related study [59]. An Arduino Uno microcontroller board was used, and a LoRa shield was used for wireless communication. The sensors measured different water quality parameters, which include pH, temperature, turbidity, and TDS. The Things Network (TTN), ThingSpeak, and ThingView were used for cloud-based data storage and visualization. Moreover, a solution for early warning of abnormal water conditions in developing countries using IoT is presented [60]. Sensors used include pH, temperature, and TDO sensors, with an ATmega328 microcontroller. ESP8266 Wi-Fi module with an external antenna is used for wireless communication. A cloud-based platform (Thinger.io) is used to store and analyze the data. Table 2 presents a summary of the solutions used in water quality monitoring

In an endeavor to enhance the capabilities of the solutions, the IoT driven solutions have also been integrated with ML for water quality prediction [47,62]. A study is presented in which a smart water quality monitoring framework uses an intelligent IoT wireless sensor network system to predict *Escherichia coli* concentrations in water using ML algorithms [63]. Different communication techniques, including LTE M, Zigbee, WiFi, Bluetooth, cellular, SigFox, LoRa, NB-IoT, and are used for sending data to the cloud.

5. Tiny machine learning (TinyML)

TinyML represents an emerging technology that shows the intersection of machine learning and embedded systems, as presented in Fig. 2.

TinyML aims to enable the deployment of ML models on resource-

Table 2
Summary IoT driven solutions for montiroing water quality.

Microcontrollers	Sensors	Connectivity
Arduino Uno [56,59]	Turbidity, Temperature,pH TDS	GSM and LoRaWAN
Arduino Board [55, 58]	Nitrate, Temperature, pH, TDO, Turbidity, TDS, Ammonia, Manganese	WiFi module, Bluetooth module
ATmega328 Microcontroller [60]	Temperature,pH, Dissolved Oxygen (DO)	ESP8266 Wi-Fi
Arduino MKR [61]	Temperature, pH, DO, Turbidity, TDS	GSM module, ESP8266 Wi-Fi, MicroSD card
Raspberry Pi 3 Model *B+* [59]	Temperature, pH, DO, Turbidity, TDS	LoRaWAN
Waspmote [57]	- Temperature, pH, DO, Turbidity, TDS, Conductivity, Oxidation-Reduction Potential (ORP)	-Bluetooth, Wi-Fi, and 4 G connectivity
Not Specified [47]		Zigbee, Wi-Fi, Bluetooth, Cellular, SigFox, LoRa, NB-IoT, LTE-M

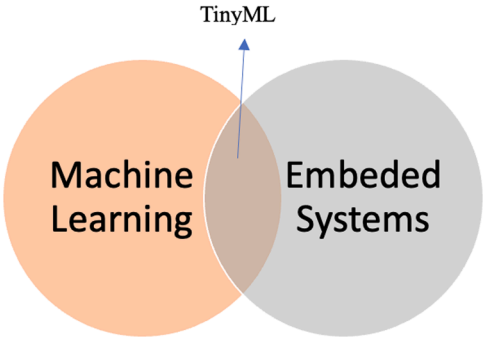


Fig. 2. A representation of TinyML.

constrained devices, like microcontrollers, that often have limited memory (mostly <1 MB) and processing power (frequently between 40 and 400 MHz) [64].This allows for real-time data processing directly on the embedded device without the need for cloud connectivity. The main objectives of TinyML include enabling real-time intelligence at the edge, reducing power consumption, minimizing latency, enhancing privacy, and reducing operational costs by localizing data processing [65]

5.1. TinyML vs traditional ML

The main difference between TinyML and traditional ML lies in the environment where the model is deployed. Traditional machine learning models require intensive computational resources (e.g., GPUs and large memory) and are deployed on high-performance systems that are accessible in cloud servers. TinyML, however, is designed to run on embedded systems with memory and compute resource constraints [64]. Traditional machine learning frameworks often prioritize model accuracy, but TinyML prioritizes model efficiency, energy consumption, and on-device processing [65] Techniques like model compression, quantization, and pruning are frequently used in TinyML to reduce the size of ML models without sacrificing much accuracy [66].

5.2. Benefits of TinyML in IoT

TinyML presents several advantages in the context of IoT [66].To begin with, TinyML models are designed for deployment on low-power microcontrollers. This enables them to be deployed in remote or battery-operated IoT devices without excessive energy consumption. Second, since TinyML allows for on-device processing, IoT devices can perform data processing in real time. This is key in avoiding the delays associated with transmitting data to the cloud. In addition, TinyML solutions make it possible to use inexpensive hardware components, such as microcontrollers, and to process data locally. This reduces the need for expenses related to cloud storage and data transmission in integrating machine learning on edge devices. Last but not least, by processing data locally on the device, TinyML minimizes the need to transmit sensitive information to the cloud increasing privacy while reducing potential vulnerabilities exploited in IoT-based solutions.

5.3. TinyML applications areas

TinyML can be applied across different domains [67]. In the healthcare sector, TinyML enables wearable devices to continuously monitor vital signs, providing early warnings for health conditions. In Smart Cities, TinyML is applied in smart city infrastructure, such as environmental monitoring, traffic control, and waste management. In agriculture, in smart agriculture, TinyML-powered solutions monitor conditions like soil moisture and crop health, optimizing resource use and improving yields. In industries, TinyML enables predictive maintenance in manufacturing, helping to prevent equipment failures by monitoring machinery for early signs of wear and tear. In environmental monitoring, TinyML sensors can be deployed to monitor water quality and air quality, detect forest fires, or track wildlife, providing valuable data for conservation and environmental management.

5.4. TinyML in water quality monitoring

Even though the application of tinyML in water quality Prediction still remains under-exploited there have been some attempts to apply tinyML water quality monitoring. A study was conducted on offline prediction of cholera in rural communal tap waters using edge AI inference [68]. IoT and AI were used to monitor water's physicochemical parameters. Data from sensors were processed on the edge to predict waterborne diseases without cloud dependency. In this study, the validation of results with traditional methods is not done, the model was not integrated into the IoT device, and real-time prediction was not possible. The study also highlighted the limited datasets available for training as a major challenge.

A tinyML approach for quantification of BOD and COD in water has also been proposed [69] Machine learning models, particularly Random Forest, were applied to minimal input parameters for predicting BOD and COD. Raspberry Pi was used for implementation. This proposed model is computationally intensive and cannot be deployed in small microcontrollers. There is also a need to validate the models. There was also limited focus on scalability and deployment in diverse geographical regions. In a related study, the design of li river water quality dynamic monitoring systems based on Raspberry Pi and TinyML was also conducted [70]. Embedded computing, deep learning, and IoT technology were combined with software and hardware design using Raspberry Pi and sensors. This study did not focus on low-power and resource-efficient models and is still highly dependent on cloud infrastructure for advanced processing, which is not a challenge in resource-constrained environments.

The reviewed studies that have attempted to apply TinyML in water quality monitoring highlight challenges such as lack of model validation with traditional methods, computational inefficiency unsuitable for deployment on low-power microcontrollers, absence of real-time prediction capabilities, and heavy reliance on cloud infrastructure. In addition, problems like limited datasets and a focus on specific geographical regions hinder the scalability and general applicability of these solutions. These limitations show the need for further research to develop validated, resource-efficient, and scalable TinyML models that can be integrated into IoT devices for real-time water quality prediction in resource-constrained environments.

6. Discussion and recommendations

6.1. Datasets

From the reviewed datasets, there is a move in water quality monitoring from the use of traditional manual sampling methods to the use of the Internet of Things (IoT) and advanced sensing technologies. This move can be attributed to the need for more efficient, continuous, and real-time data collection. The sensors used in these solutions can measure the physiochemical properties of water, which include temperature, pH, electrical conductivity, dissolved oxygen, and turbidity. This move towards IoT and sensing technologies has been driven by challenges associated with traditional monitoring methods. In addition, accessing remote or hazardous locations for sampling presents safety risks and difficulit to access. IoT devices and sensors overcome these challenges by enabling remote, automated, and real-time monitoring.

Even though global water quality datasets collected using IoT and sensing technologies are available, there is inadequate data specific to Kenyan water bodies. Most existing datasets are from other regions in the world with a limited focus on Africa. This may mean they do not capture the unique environmental conditions, pollution sources, and socio-economic factors influencing water quality in Africa, specifically in Kenya. The limited localized data has hindered the understanding of Kenya's water quality problems. Therefore, collecting data in Kenya is essential to obtain accurate and relevant insights that can inform effective water management and policy decisions tailored to the needs of the country and other developing countries. In addition, the technological environment and infrastructure in Kenya require special considerations when developing IoT and sensing technologies. Existing studies do not address the challenges, which include network connectivity in rural areas, energy requirements, and model maintenance in the Kenyan setting. This can affect the performance and feasibility of these technologies. A localized approach is needed that not only fills a data gap but also contributes to building local capacity and enhancing global understanding of water quality issues in developing countries.

6.2. ML models

Machine learning is indeed an important tool in analyzing and predicting water quality. Various ML algorithms have demonstrated high accuracies across different studies. Among the most frequently used models are Random Forest (RF), Support Vector Machine (SVM), Artificial Neural Networks (ANN), and gradient boosting algorithms like LightGBM and XGBoost. Random Forest has been specifically highlighted in several studies for its better performance in predicting water quality parameters such as pH, turbidity, dissolved oxygen, and conductivity. In addition, SVMs have shown strong predictive capabilities when classifying water as safe or unsafe for domestic use. Gradient boosting models like LightGBM and XGBoost have also demonstrated high accuracies, making them effective for water quality classification tasks. Artificial Neural Networks, including Multi-Layer Perceptron (MLP) models, have been effective in capturing complex nonlinear relationships in environmental data in predicting the WQI. Other algorithms, including Decision Trees, K-Nearest Neighbors (KNN), and ensemble methods, for example, AdaBoost and Gradient Boosting have also been utilized. However, their effectiveness often depends on the specific parameters and datasets involved.

The limitation of these studies that developed ML models for water quality monitoring was the use of a single dataset. The use of single datasets may limit the generalizability of the findings. There was also a need for further optimization of some of the machine learning algorithms, suggesting that the current results may not represent the best possible performance. In addition, the use of a publicly available dataset may have characteristics specific to the source and limit the ability to generalize the findings to other water quality datasets.

There is a need to expand the dataset in future works by collecting additional data from multiple sources, regions, and periods. This will lead to the creation of a more diverse and representative sample of water quality data. The collected data can further be integrated with publicly available datasets, increasing the variability and richness of the water quality dataset. In addition, there is a need to optimize the machine learning algorithms to achieve the best possible performance. Optimization can involve hyperparameter tuning using optimization techniques like grid search, random search, or Bayesian optimization. Experimenting should also be done with a variety of machine learning algorithms while enhancing datasets through feature selection and extraction methods to improve model accuracy. To enhance generalization domain adaptation techniques can be applied to standardize data collection and processing methods across different sources. External validation should also be performed on independent datasets not used during training to assess model performance in different contexts.

6.3. Water quality monitoring

In areas with connectivity and resource constraints, such as many regions in Africa, IoT-based water quality monitoring systems face different challenges. To begin with, limited internet connectivity interferes with real-time data transmission to cloud servers where data is usually stored and analyzed, energy problems, including unreliable power supplies, affect continuous operation and high operational costs which may include expenses for data plans and maintenance. In addition, a shortage of technical expertise complicates the installation and maintenance of the solutions. These factors have made it difficult to scale and sustain traditional IoT solutions in such settings.

To address these challenges, innovations like TinyML are needed. TinyML can enable on-device data processing on low-power microcontrollers. This reduces reliance on constant connectivity and lowers energy consumption. By analyzing data locally, systems can provide real-time insights and alerts without frequent data transmission. This will minimize bandwidth use and operational costs of the solutions. Integrating TinyML into water quality monitoring offers a sustainable and effective approach tailored to resource-constrained environments and can enhance scalability and responsiveness while mitigating the challenges of limited connectivity and resources. In the next section, TinyML is presented in detail.

6.4. Conceptual model for water quality monitoring

A conceptual model that can guide the development of a model for real-time water quality prediction in resource-constrained environments is proposed as presented in Fig. 3. The key variables involved are as follows:

1. Independent Variables:
   Physicochemical Properties of water: These include measurable factors such as pH, turbidity, dissolved oxygen, conductivity, and temperature. They are the inputs collected from water sources using IoT sensors.
2. Mediating Variable:
   Machine Learning Models: The ML models process the water quality parameters to predict the water quality status. They mediate the relationship between the independent variables and the dependent variable by analyzing the input data to generate predictions.
3. Moderating Variables:
   Resource Constraints: These factors influence how effectively the ML models can process data and generate predictions. They moderate the relationship between the ML models and the time and accuracy of water quality prediction.
   - Energy challenges affect the duration devices can stay on and the processing capabilities of devices.
   - Connectivity impacts data transmission to and from cloud-based models.

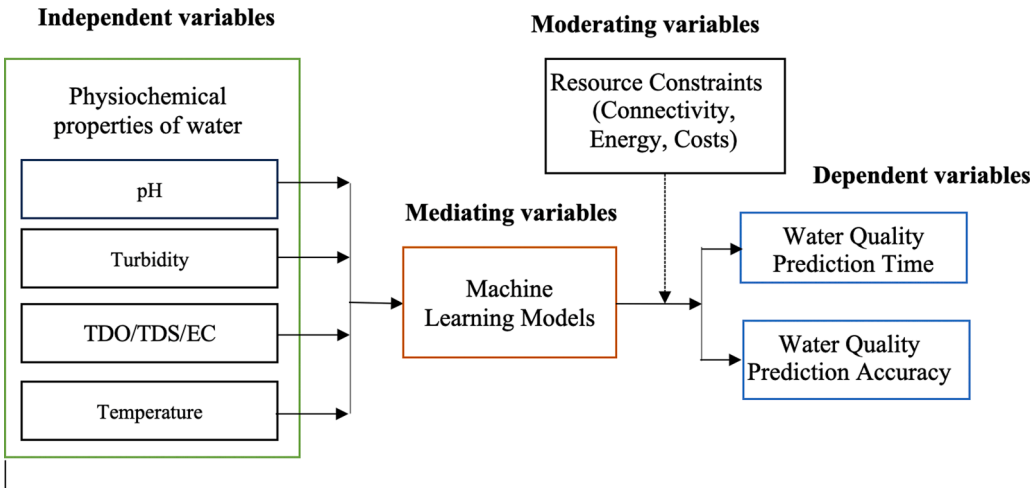


Fig. 3. Conceptual model for Realtime water quality monitoring.

- The processing power determines the complexity of ML models that can be run on edge devices.

4. Dependent Variable:
   Water Quality Prediction time and accuracy: The outcome of the system, representing the ability to accurately predict water quality in real-time based on the input parameters and processing by the ML models.

7. Conclusion

This review highlights the potential of TinyML and IoT for real-time, scalable water quality monitoring in developing countries. Unlike previous reviews, this study focuses on the feasibility of on-device ML processing, identifying key challenges in dataset availability and model validation. Addressing these gaps will be crucial for developing low-cost, deployable solutions that support sustainable water management and public health. This addresses the limitations of traditional methods and cloud-based models. Despite efforts to apply the latest technologies, several challenges are still experienced, including the lack of localized datasets, the need for model validation, energy challenges, and connectivity problems. Addressing these gaps requires the development of a framework that incorporates localized data collection, model optimization, and deployment of TinyML for edge computing. By using localized data and validating models against traditional methods, future research can enhance the accuracy and relevance of predictive models for water quality monitoring. This approach can significantly contribute to sustainable water management, supporting global health and environmental goals.

8. Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Elicit to analyze the abstracts of the reviewed papers to determine relevance. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

CRediT authorship contribution statement

Samson Otieno Ooko: Writing – original draft, Methodology, Conceptualization, Writing – review & editing, Visualization, Investigation. **Laura Cheptegei:** Validation, Conceptualization, Writing – review & editing, Supervision. **Simon M. Karume:** Writing – review & editing, Validation, Conceptualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] S. Perveen, Amar-Ul-Haque, Drinking Water Quality monitoring, Assessment and Management in Pakistan: A Review, Elsevier Ltd, 2023, https://doi.org/10.1016/j.heliyon.2023.e13872.
[2] D.K. Lukhabi, P.K. Mensah, N.K. Asare, T. Pulumuka-Kamanga, K.O. Ouma, Adapted Water Quality Indices: Limitations and Potential for Water Quality Monitoring in Africa, MDPI, 2023, https://doi.org/10.3390/w15091736.
[3] K.B. Githaiga, S.M. Njuguna, R.W. Gituru, X. Yan, Water quality assessment, multivariate analysis and human health risks of heavy metals in eight major lakes in Kenya, J. Environ. Manage. 297 (2021), https://doi.org/10.1016/j.jenvman.2021.113410.
[4] M. Ngatia, S.M. Kithiia, M. Voda, Effects of anthropogenic activities on water quality within Ngong River Sub-catchment, Nairobi, Kenya, Water (Switzerland) 15 (4) (2023), https://doi.org/10.3390/w15040660.
[5] N.T. Anh, L.D. Can, N.T. Nhan, B. Schmalz, T.Le Luu, Influences of key factors on river water quality in urban and rural areas: a review, Case Stud. Chem. Environ. Eng. 8 (2023), https://doi.org/10.1016/j.cscee.2023.100424.
[6] P. Chen, B. Wang, Y. Wu, Q. Wang, Z. Huang, C. Wang, Urban river water quality monitoring based on self-optimizing machine learning method using multi-source remote sensing data, Ecol. Indic. 146 (2023), https://doi.org/10.1016/j.ecolind.2022.109750.
[7] P. Lindholm-Lehto, Water quality monitoring in recirculating aquaculture systems, Aquac., Fish Fisheries 3 (2) (2023) 113–131, https://doi.org/10.1002/aff2.102.
[8] M. Miller, A. Kisiel, D. Cembrowska-Lech, I. Durlik, T. Miller, IoT in water quality monitoring—Are we really here? Sensors 23 (2) (2023) https://doi.org/10.3390/s23020960.
[9] C. Ma, J. Zhao, B. Ai, S. Sun, Z. Yang, Machine learning based long-term water quality in the turbid Pearl River Estuary, China, J. Geophys. Res. Oceans 127 (1) (2022) 1–19, https://doi.org/10.1029/2021JC018017.
[10] H. Bamoumen, A. Temouden, N. Benamar, Y. Chtouki, How TinyML can be leveraged to solve environmental problems: a survey, in: 2022 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies, 3ICT 2022, Institute of Electrical and Electronics Engineers Inc., 2022, pp. 338–343, https://doi.org/10.1109/3ICT56508.2022.9990661.
[11] C.R. Banbury et al., "Benchmarking TinyML systems: challenges and direction," 2020, [Online]. Available: http://arxiv.org/abs/2003.04821.
[12] P. Jayaraman, K.K. Nagarajan, P. Partheeban, V. Krishnamurthy, Critical review on water quality analysis using IoT and machine learning models, Int. J. Inf. Manag. Data Insights 4 (1) (2024), https://doi.org/10.1016/j.jjimei.2023.100210.
[13] J. Zhang, Y. Sheng, W. Chen, H. Lin, G. Sun, P. Guo, Design and analysis of a water quality monitoring data service platform, Comput., Mater. Continua 66 (1) (2021) 389–405, https://doi.org/10.32604/cmc.2020.012384.
[14] M. Maiolo, D. Pantusa, Multivariate analysis of water quality data for drinking water supply systems, Water (Switzerland) 13 (13) (2021), https://doi.org/10.3390/w13131766.

[15] H.K. Namatovu, M.A. Magumba, T.J. Oyana, A water quality dataset of levels of metal, nutrient and anions in sample water points from sixteen selected urban and rural districts of Uganda, Data Brief 50 (2023), https://doi.org/10.1016/j.dib.2023.109601.

[16] A.S. Ejoh, B.A. Unuakpa, F.H. Ibadin, S.O. Edeki, Dataset on the assessment of water quality and water quality index of Ubogo and Egini rivers, Udu LGA, Delta State Nigeria, Data Brief 19 (2018) 1716–1726, https://doi.org/10.1016/j.dib.2018.06.053.

[17] R. Kumar, A. Raina, N. Sharma, Dataset on water quality characteristics of a hill stream in Bhaderwah, Jammu and Kashmir, Data Brief 26 (2019), https://doi.org/10.1016/j.dib.2019.104462.

[18] B. Srivastava, S.S. Sandha, V. Raychoudhury, S. Randhawa, V. Kapoor, A. Agrawal. 2020 IEEE International Conference on Pervasive Computing and Communications Workshops, Building an Open, Multi-Sensor, Dataset of Water Pollution of Ganga Basin and Application to Assess Impact of Large Religious Gatherings, PerCom Workshops, Austin, TX, USA, 2020, pp. 1–6. https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=9156149&isnumber=9156071.

[19] F. Ahmedi, L. Ahmedi, Dataset on water quality monitoring from a wireless sensor network in a river in Kosovo, Data Brief 44 (2022), https://doi.org/10.1016/j.dib.2022.108486.

[20] A. Schubert, J. Harrison, L. Kent-Buchanan, V. Bonds, S.P. McElmurry, N.G. Love, A point-of-use drinking water quality dataset from fieldwork in Detroit, Michigan, Sci. Data 11 (1) (2024), https://doi.org/10.1038/s41597-024-03298-w.

[21] J. Krohkaew, P. Nilaphruek, N. Witthayawiroj, S. Uapipatanakul, Y. Thwe, P. N. Crisnapati, Thailand raw water quality dataset analysis and evaluation, Data (Basel) 8 (9) (2023), https://doi.org/10.3390/data8090141.

[22] S. Shakhari, I. Banerjee, Clustering-based water quality exploration and visualization in municipal water supply, in: 2024 IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation, IATMSI 2024, Institute of Electrical and Electronics Engineers Inc., 2024, https://doi.org/10.1109/IATMSI60426.2024.10503581.

[23] M.P.P. Sithole, N.I. Nwulu, E.M. Dogo, Dataset for a wireless sensor network based drinking-water quality monitoring and notification system, Data Brief 27 (2019), https://doi.org/10.1016/j.dib.2019.104813.

[24] A.I. Arafat, T. Akter, M.F. Ahammed, M.Y. Ali, A. Al Nahid, A dataset for internet of things based fish farm monitoring and notification system, Data Brief 33 (2020), https://doi.org/10.1016/j.dib.2020.106457.

[25] C.N. Udanor, et al., An internet of things labelled dataset for aquaponics fish pond water quality monitoring system, Data Brief 43 (2022), https://doi.org/10.1016/j.dib.2022.108400.

[26] A. Castellini, D. Bloisi, J. Blum, F. Masillo, A. Farinelli, Multivariate sensor signals collected by aquatic drones involved in water monitoring: a complete dataset, Data Brief 30 (2020), https://doi.org/10.1016/j.dib.2020.105436.

[27] M.R.V. Ross, et al., AquaSat: a data set to enable remote sensing of water quality for inland waters, Water Resour. Res. 55 (11) (2019) 10012–10025, https://doi.org/10.1029/2019WR024883.

[28] V. Sagan, et al., Monitoring Inland Water Quality Using Remote sensing: Potential and Limitations of Spectral indices, Bio-Optical simulations, Machine learning, and Cloud Computing, Elsevier B.V., 2020, https://doi.org/10.1016/j.earscirev.2020.103187.

[29] H. Virro, G. Amatulli, A. Kmoch, L. Shen, E. Uuemaa, GRQA: global River Water Quality Archive, Earth Syst Sci Data 13 (12) (2021) 5483–5507, https://doi.org/10.5194/essd-13-5483-2021.

[30] M. Zhu, et al., A Review of the Application of Machine Learning in Water Quality Evaluation, Elsevier B.V., 2022, https://doi.org/10.1016/j.eehl.2022.06.001.

[31] S. Y, S. Amith, S. S, Dr.A.S. Sastry, Monitoring of water quality using machine learning - A review, Int. J. Res. Appl. Sci. Eng. Technol. 12 (5) (2024) 495–498, https://doi.org/10.22214/ijraset.2024.61578.

[32] N.S. Pagadala, M. Marri, A. Myla, B. Abburi, K.S. Ramtej, Water quality prediction using machine learning techniques, in: Proceedings of the 10th International Conference on Signal Processing and Integrated Networks, SPIN 2023, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 358–362, https://doi.org/10.1109/SPIN57001.2023.10117415.

[33] S.P. Shingare, M.B. Mandake, R.N. Patil, A.G. Thokal, Y. Qureshi, C.M. Thakar, Applicability of machine learning in waste water quality detection, in: Proceedings - International Conference on Computing, Power, and Communication Technologies, IC2PCT 2024, Institute of Electrical and Electronics Engineers Inc., 2024, pp. 525–530, https://doi.org/10.1109/IC2PCT60090.2024.10486814.

[34] S. Kaddoura, Evaluation of machine learning algorithm on drinking water quality for better sustainability, Sustain. (Switzerland) 14 (18) (2022), https://doi.org/10.3390/su141811478.

[35] G.S… Tomar, in: Proceedings, 2023 IEEE 12th International Conference on Communication Systems and Network Technologies (CSNT 2023), Madhya Pradesh Section Flagship Conference, IEEE, 2023.

[36] R. Akshay, G. Tarun, P.U. Kiran, K.D. Devi, M. Vidhyalakshmi, Water-quality-analysis using machine Learning, in: Proceedings of the 2022 11th International Conference on System Modeling and Advancement in Research Trends, SMART 2022, Institute of Electrical and Electronics Engineers Inc., 2022, pp. 13–18, https://doi.org/10.1109/SMART55829.2022.10047533.

[37] C. Saab, G.P. Zehil, About machine learning techniques in water quality monitoring, in: 2023 5th International Conference on Advances in Computational Tools for Engineering Applications, ACTEA 2023, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 115–121, https://doi.org/10.1109/ACTEA58025.2023.10193911.

[38] S.Y. Abuzir, Y.S. Abuzir, Machine learning for water quality classification, Water Quality Res. J. 57 (3) (2022) 152–164, https://doi.org/10.2166/wqrj.2022.004.

[39] N. Wagle, T.D. Acharya, D.H. Lee, Comprehensive Review on Application of Machine Learning Algorithms for Water Quality Parameter Estimation Using Remote Sensing Data, M Y U Scientific Publishing Division, 2020, https://doi.org/10.18494/SAM.2020.2953.

[40] D. Brindha, V. Puli, N.V.S.S. Bala Karthik Sobula, V.S. Mittakandala, G. D. Nanneboina, Water quality analysis and prediction using machine Learning, in: Proceedings - 7th International Conference on Computing Methodologies and Communication, ICCMC 2023, Institute of Electrical and Electronics Engineers Inc., 2023, pp. 175–180, https://doi.org/10.1109/ICCMC56507.2023.10083776.

[41] S.O. Ooko, E.K. Pamela, G. Kwagalakwe, Use of machine learning for realtime water quality prediction, in: IEEE AFRICON Conference, Institute of Electrical and Electronics Engineers Inc., 2023, https://doi.org/10.1109/AFRICON55910.2023.10293701.

[42] A. Kuthe, C. Bhake, V. Bhoyar, A. Yenurkar, V. Khandekar, K. Gawale, Water quality analysis using machine learning, Int. J. Res. Appl. Sci. Eng. Technol. 10 (12) (2022) 581–585, https://doi.org/10.22214/ijraset.2022.47944.

[43] M.Y. Shams, A.M. Elshewey, E.S.M. El-kenawy, A. Ibrahim, F.M. Talaat, Z. Tarek, Water quality prediction using machine learning models based on grid search method, Multimed Tools Appl. 83 (12) (2024) 35307–35334, https://doi.org/10.1007/s11042-023-16737-4.

[44] E. Dritsas, M. Trigka, Efficient data-driven machine learning models for water quality prediction, Computation 11 (2) (2023), https://doi.org/10.3390/computation11020016.

[45] O.K. Pal, The quality of drinkable water using machine learning techniques, Int. J. Adv. Eng. Res. Sci. 9 (6) (2022) 016–023, https://doi.org/10.22161/ijaers.96.2.

[46] C. Varadharajan, et al., Can machine learning accelerate process understanding and decision-relevant predictions of river water quality? Hydrol Process 36 (4) (2022) https://doi.org/10.1002/hyp.14565.

[47] M.A. Rahu, A.F. Chandio, K. Aurangzeb, S. Karim, M. Alhussein, M.S. Anwar, Toward design of internet of Things and machine learning-enabled frameworks for analysis and prediction of water quality, IEEE Access 11 (2023) 101055–101086, https://doi.org/10.1109/ACCESS.2023.3315649.

[48] A. El Bilali, A. Taleb, Prediction of irrigation water quality parameters using machine learning models in a semi-arid environment, J. Saudi Society Agric. Sci. 19 (7) (2020) 439–451, https://doi.org/10.1016/j.jssas.2020.08.001.

[49] Md.M. Hassan, et al., Efficient prediction of water quality index (WQI) using machine learning algorithms, Human-Centric Intell. Systems 1 (3–4) (2021) 86, https://doi.org/10.2991/hcis.k.211203.001.

[50] P. Dandekar, V. Thakre, K. Sharma, A. Kushwaha, A predictive model for water quality index assessment by machine learning approach, in: 2024 2nd International Conference on Computer, Communication and Control, IC4, Institute of Electrical and Electronics Engineers Inc., 2024, p. 2024, https://doi.org/10.1109/IC457434.2024.10486535.

[51] S.S. Mukonza, J.L. Chiang, Meta-Analysis of Satellite Observations for United Nations Sustainable Development Goals: Exploring the Potential of Machine Learning for Water Quality Monitoring, Multidisciplinary Digital Publishing Institute (MDPI), 2023, https://doi.org/10.3390/environments10100170.

[52] N. Hassan, C.S. Woo, Machine learning application in water quality using satellite data, in: IOP Conference Series: Earth and Environmental Science, IOP Publishing Ltd, 2021, https://doi.org/10.1088/1755-1315/842/1/012018.

[53] A. Thakur, P. Devi, A Comprehensive Review On Water Quality Monitoring Devices: Materials Advances, Current Status, and Future Perspective, Taylor and Francis Ltd, 2024, https://doi.org/10.1080/10408347.2022.2070838.

[54] E.T. de Camargo, et al., Low-Cost Water Quality Sensors for IoT: A Systematic Review, MDPI, 2023, https://doi.org/10.3390/s23094424.

[55] P.G. Arepalli, K.J. Naik, An IoT based smart water quality assessment framework for aqua-ponds management using Dilated spatial-temporal Convolution Neural Network (DSTCNN), Aquac. Eng. 104 (2024), https://doi.org/10.1016/j.aquaeng.2023.102373.

[56] R.P. Shete, A.M. Bongale, D. Dharrao, IoT-enabled effective real-time water quality monitoring method for aquaculture, MethodsX 13 (2024), https://doi.org/10.1016/j.mex.2024.102906.

[57] M. Kumar, T. Singh, M.K. Maurya, A. Shivhare, A. Raut, P.K. Singh, Quality assessment and monitoring of river water using IoT infrastructure, IEEE Internet Things J 10 (12) (2023) 10280–10290, https://doi.org/10.1109/JIOT.2023.3238123.

[58] R. Bogdan, C. Paliuc, M. Crisan-Vida, S. Nimara, D. Barmayoun, Low-cost internet-of-things water-quality monitoring system for rural areas, Sensors 23 (8) (2023), https://doi.org/10.3390/s23083919.

[59] W.A. Jabbar, et al., Development of LoRaWAN-based IoT system for water quality monitoring in rural areas, Expert Syst. Appl. 242 (2024), https://doi.org/10.1016/j.eswa.2023.122862.

[60] O.O. Flores-Cortez, A low-cost IoT system for water quality monitoring in developing countries, in: Proceedings - IEEE Consumer Communications and Networking Conference, CCNC, Institute of Electrical and Electronics Engineers Inc, 2024, pp. 596–597, https://doi.org/10.1109/CCNC51664.2024.10454847.

[61] K. Lal, S. Menon, F. Noble, K.M. Arif, Low-cost IoT based system for lake water quality monitoring, PLoS One 19 (2024), https://doi.org/10.1371/journal.pone.0299089, 3 March.

[62] N. Rawat, Water quality prediction using machine learning, Int. J. Res. Appl. Sci. Eng. Technol. 10 (6) (2022) 4173–4187, https://doi.org/10.22214/ijraset.2022.44658.

[63] Y. Singh, T. Walingo, Smart water quality monitoring with IoT wireless sensor networks, Sensors 24 (9) (2024), https://doi.org/10.3390/s24092871.

[64] A. Karras, et al., TinyML algorithms for big data management in large-scale IoT systems, Future Int. 16 (2) (2024), https://doi.org/10.3390/fi16020042.

[65] L. Capogrosso, F. Cunico, D.S. Cheng, F. Fummi, M. Cristani, A machine learning-oriented survey on tiny machine learning, IEEE Access 12 (2024) 23406–23426, https://doi.org/10.1109/ACCESS.2024.3365349.
[66] N.N. Alajlan, D.M. Ibrahim, TinyML: adopting tiny machine learning in smart cities, J. Autonomous Intell. 7 (4) (2024), https://doi.org/10.32629/jai.v7i4.1186.
[67] A. Elhanashi, P. Dini, S. Saponara, Q. Zheng, Advancements in TinyML: Applications, Limitations, and Impact on IoT Devices, Multidisciplinary Digital Publishing Institute (MDPI), 2024, https://doi.org/10.3390/electronics13173562.
[68] M.M. Ogore, K. Nkurikiyeyezu, J. Nsenga, Offline prediction of cholera in rural communal tap waters using edge AI inference, in: 2021 IEEE Globecom Workshops, GC Wkshps 2021 - Proceedings, Institute of Electrical and Electronics Engineers Inc., 2021, https://doi.org/10.1109/GCWkshps52748.2021.9682128.
[69] S. Soni, A. Khurshid, A.M. Minase, A. Bonkinpelliwar, A TinyML approach for quantification of BOD and COD in water, in: 2023 2nd International Conference on Paradigm Shifts in Communications Embedded Systems, Machine Learning and Signal Processing, PCEMS 2023, Institute of Electrical and Electronics Engineers Inc., 2023, https://doi.org/10.1109/PCEMS58491.2023.10136050.
[70] X. Tang, Design of Li River Water Quality Dynamic monitoring system based on Raspberry Pi and TinyML, J. Comput. Commun. 11 (2023) 121–133, https://doi.org/10.4236/jcc.2023.118009.